

Bachelor Thesis

Static Prompt Injection Detection via Abstract Meaning Representation

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Abstract

Prompt injection attacks cause LLM agents to execute malicious instructions embedded in user inputs or tool outputs, overriding system policy. Existing defenses rely on keyword filters or LLM judges; both operate on surface text and are either trivially bypassed or themselves injection-vulnerable.

This thesis investigates whether Abstract Meaning Representation (AMR) can serve as the basis for static prompt injection detection. AMR collapses text into canonical predicate-argument graphs, discarding syntactic variation and surface obfuscation while preserving semantic intent, making static policy checking tractable.

We propose a scoped formal definition of prompt injection as a payload whose AMR graph semantically contradicts an explicit policy AMR graph. The detection mechanism combines static structural rules (polarity inversion, antonym and synonym predicates) with minimal LLM sub-calls scoped to single-concept classification queries, a sub-task too narrow for an attacker to exploit via injection. Evaluated on a custom attack suite built on the AgentDojo banking benchmark, the primary configuration achieves high detection rates with zero false positives, substantially reducing missed attacks over a no-firewall baseline. Smatch, an existing AMR graph-similarity metric explored as an alternative, is rejected because concept-agnostic matching cannot distinguish malicious instructions from legitimate transaction records. Remaining misses are tied to AMR parse quality: obfuscation techniques such as paraphrasing or indirect phrasing can prevent a detection rule from firing.

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